nonzero vectors in R^3, with v not a multiple of u, then Span{u,v} is the plane in R^3 that contains u, v and 0.

But if v is a multiple of u, then v lies on Span{u} which is the set of points on the line in R^3 through u and 0.

$\therefore$ The given statement can be either True or False depending on whether v is a multiple of u or not.

---

24) a) Any list of five real numbers is a vector in R^5.

$\boxed{\text{Solution:}}$ On Page 27 under the section "Vectors in R^n", if n is a positive integer, R^n denotes the collection of all lists (or ordered n-tuples) of n real numbers, usually written as $n \times 1$ column matrices, such as

$$u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}.$$

$\therefore$ when $n=5$, Any list of 5 real numbers is a vector in R^5. So, the statement is True. II

b) The vector u results when the vector $(u-v)$ is added to the vector v.

$\boxed{\text{Solution:}}$ The statement is True. We shall prove this by using the Algebraic Properties of R^n mentioned in Page 27.

$(u-v)+v = (u+(-v))+v$ (Notation simplicity on Pg 27)

$= u+((-v)+v)$ (Property (ii))

$= u+0$ (Property (iv)) $= u$ (Property (iii))

c) The weights $c_1, \ldots, c_p$ in a linear combination $c_1v_1 + \ldots + c_pv_p$ cannot all be zero.

$\boxed{\text{Solution:}}$ On Page 28, just above Example 4, it is mentioned that the weights in a linear combination can be any real numbers, including zero.

So, the statement is False. II

d) When u and v are nonzero vectors, Span{u,v} contains the line through u and the origin.

$\boxed{\text{Solution:}}$ This statement is True. It is mentioned on the last sentence of the section "A Geometric Description of Span{v} and Span{u,v}" on Page 30.